ACEFA
AUSTRALIA-AOTEAROA CONSORTIUM FOR EPIDEMIC FORECASTING & ANALYTICS

Lecture 5
Analysis of epidemic time-series part 2: biases in the observation process

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What is observed?



- In the last lecture we described how to quantify trends in a time series of infections
- The underlying time series of infections are often not known
  - Infections are not directly observable
- Instead, surveillance systems monitor other downstream quantities:
  - Cases
  - Hospitalisations
  - Deaths
  - Viral concentration in wastewater
  - The test-positive proportion (more on this in lecture 6)
- We observe these quantities as we might expect them to correlate with, and be proportional to the infection time series

Features that arise from surveillance systems

- Time series data collected through infectious disease surveillance systems are imperfect due to, for example:
  - Noise
  - Reporting delays
  - Delays to measurable outcomes
  - Day-of-the-week effects
  - Truncation
  - Censoring
  - Missing data
  - Changes in behaviour (e.g. testing, test-seeking, healthcare-seeking) (more on this in lecture 6)

This lecture's scope



In this lecture we will:

- Describe common data problems arising from the *observation processes* of surveillance systems
  - The focus will be on noise and delays
- Provide an overview of the methods that exist for accounting for these data problems
  - You will gain experience implementing these methods in this afternoon's practical
- Emphasise the importance of maintaining data integrity so that such methods can be used

We will expand on other biases in surveillance systems in lecture 6

Noise in epidemic time series

Sources of noise

- There can be many sources of noise or uncertainty in observation processes, e.g.
  - Fundamental noise
  - Day-of-the-week effects
  - Reporting errors
  - Missing periods of data
  - Missing fields of data

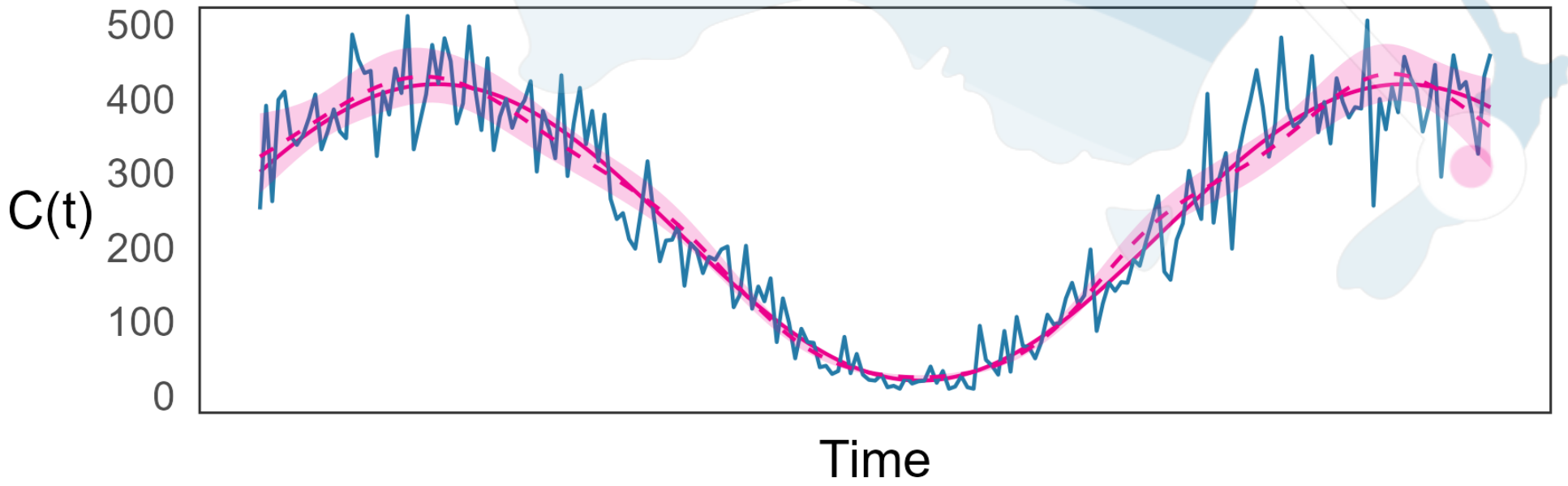
- Statistical methods can handle noise appropriately

- As epidemiologists, it is important to:
  - Recognise sources of noise
  - Employ statistical methods for dealing with noise (where possible)
  - Understand biases that may be present if unable to account for a source of noise

- Data integrity is crucial for minimising noise and uncertainty, and allowing appropriate statistical methods to be employed

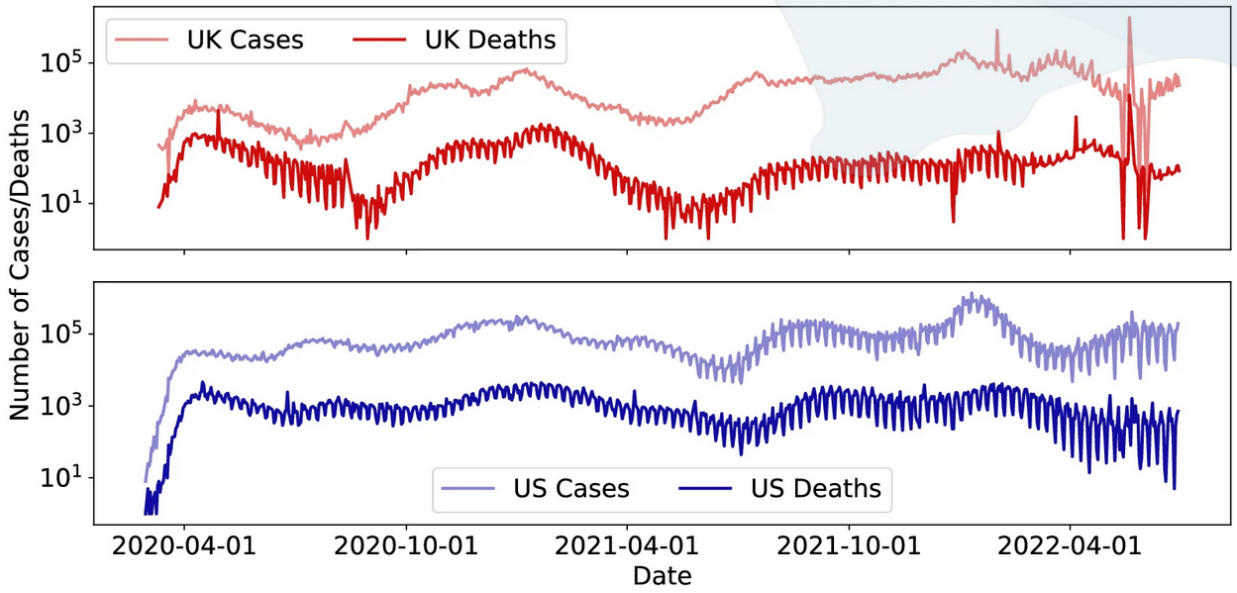
Fundamental noise

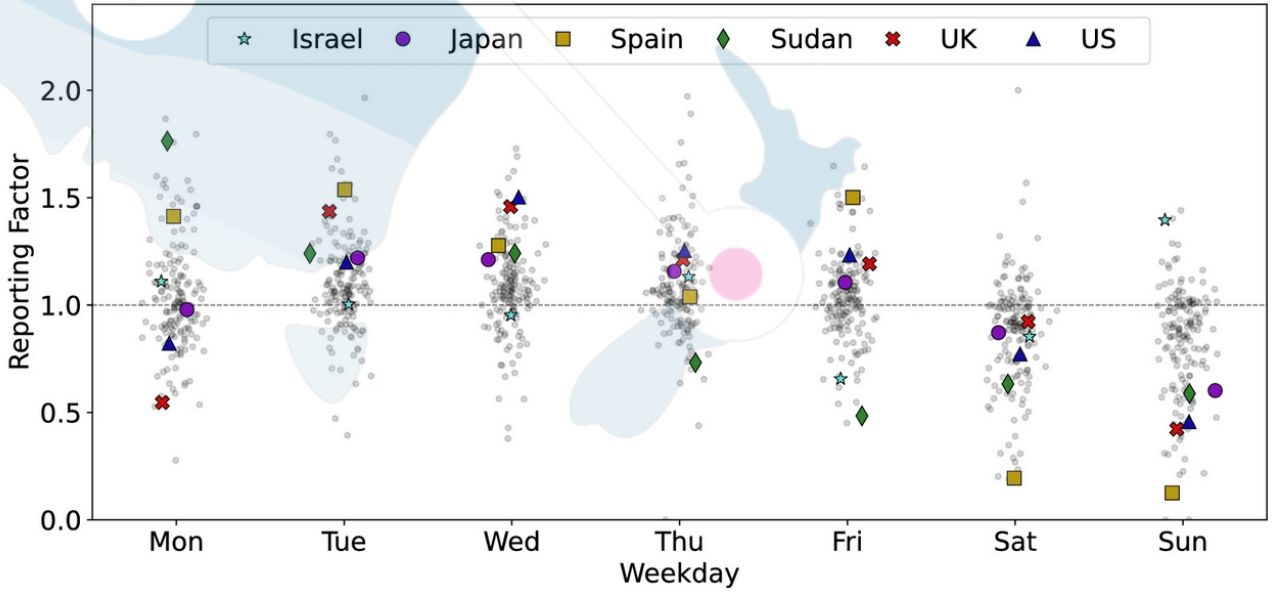
- Any observation process will include noise in the observation/event.
- Our aim is often to quantify the underlying trends in a time-series (i.e. the expected number of events through time).
- Statistical methods can be used to quantify the underlying trends in an epidemic time-series, accounting for fundamental noise in the observation process.



Day-of-the-week effects

- Many epidemic time-series exhibit day-of-the-week effects
  - Specific days of the week will be consistently higher/lower
- Day-of-the-week effects do not (usually) reflect epidemic dynamics
  - Reflect a bias in the observation process



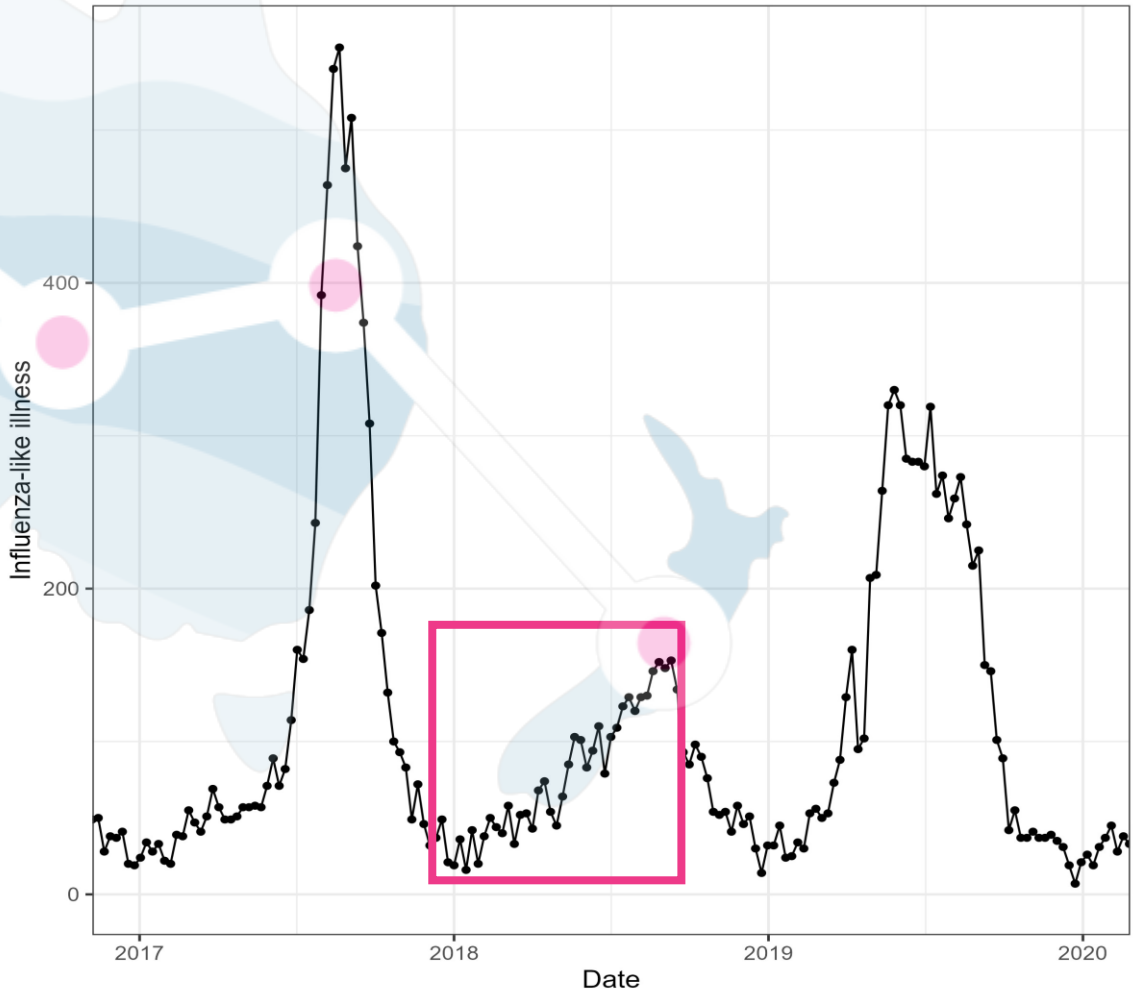


Gallagher et al. 2025. BMC Research Notes

Accounting for noise

Why not aggregate into weekly data?

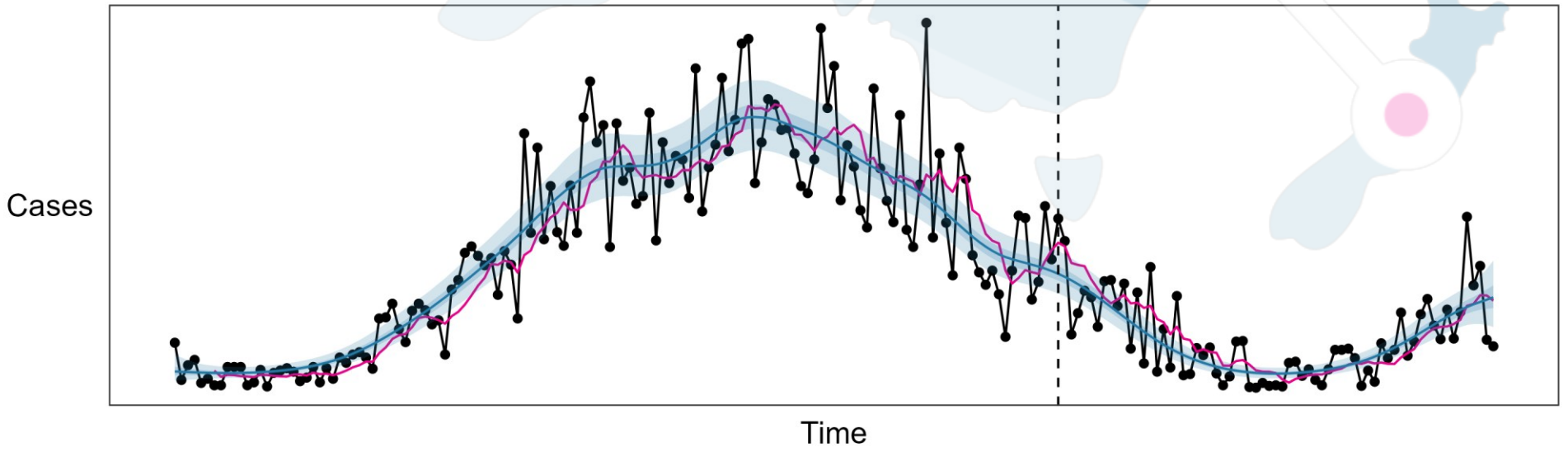
- Aggregating data can improve visualisation but throws away information
- There can still be substantial noise in e.g. weekly epidemic time-series



Why not a seven-day rolling average?

- Rolling averages can remove the appearance of some noise, but they throw out information
- Do not capture uncertainty in trends

- Better to use a statistical approach
- Statistical models can be used to separate meaningful trends in the data from noise

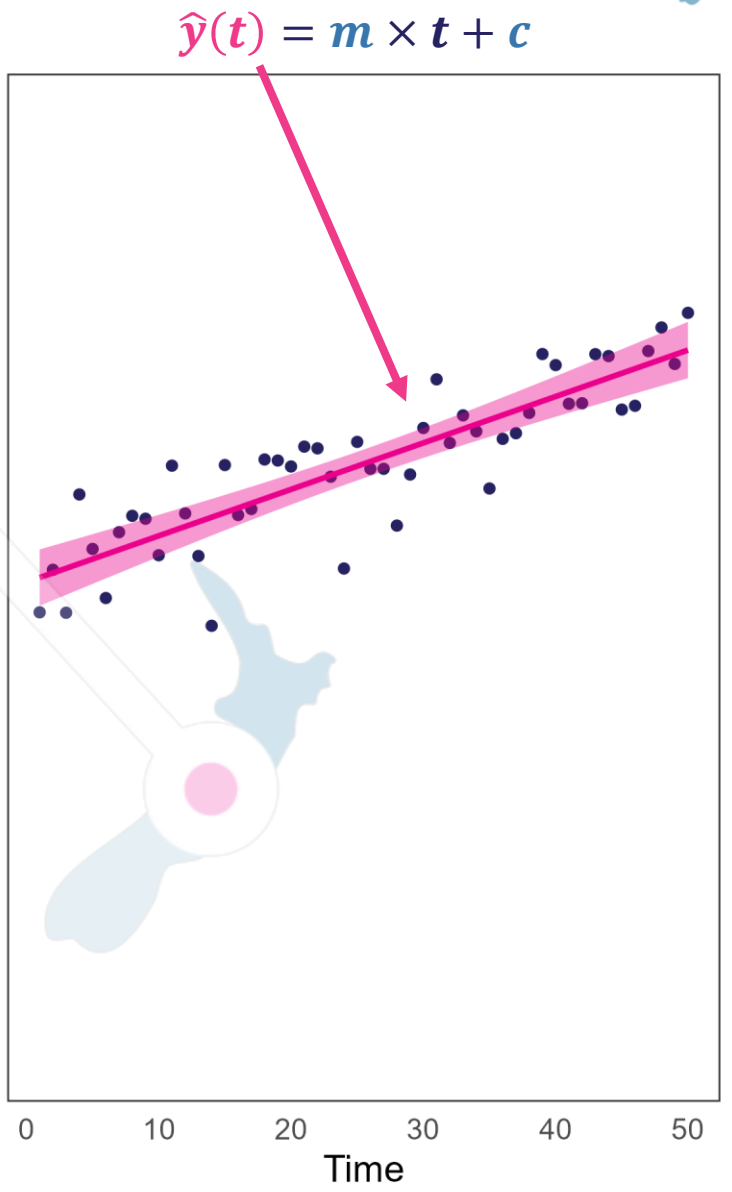


Statistical approaches

- Statistical models define a mathematical relationship between variables and data, accounting for the data generating process
  - For time series data we are interested in the relationship between time, t, and the time series data, $y(t)$
- The mathematical relationship/function with have some parameters, θ_i, that describe the relationship between time and the modelled value of the time series data: $\hat{y}(t) = f(t, \theta_i)$
  - Our goal is to determine the values and uncertainty of the parameters, θ_i
- The mathematical relationship/function could be, for example:
  - A straight line (e.g. intercept and gradient)
  - An SIR-type model (e.g. beta and gamma)
  - A general smooth model such as “splines” or a “random-walk”

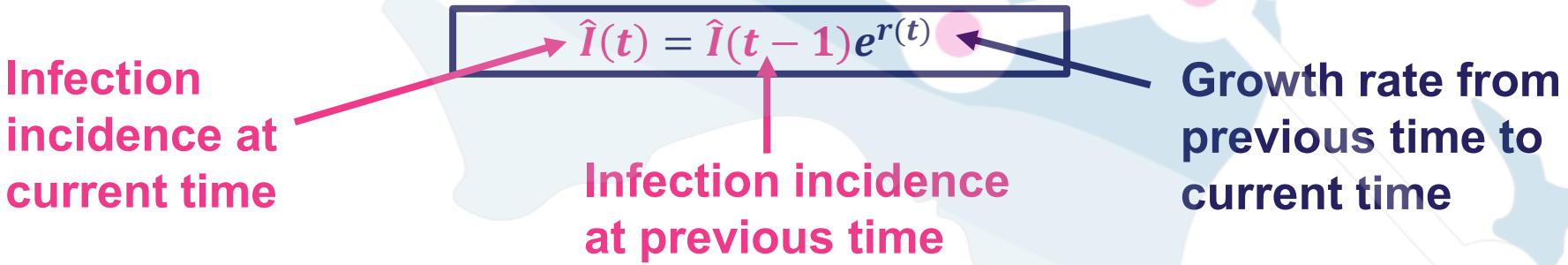
Statistical approaches

- We have:
  - Time series data $y(t)$
  - A function to derive a modelled time-series $\hat{y}(t) = f(t, \theta_i)$
- We then want to estimate the value and uncertainty in the parameters θ_i
- We define the probability (likelihood) of observing the data given values of parameters θ_i
- Methods for estimating the values and uncertainty in θ_i and therefore $\hat{y}(t)$ include:
  - Maximum likelihood approaches
  - Bayesian approaches (e.g. MCMC)



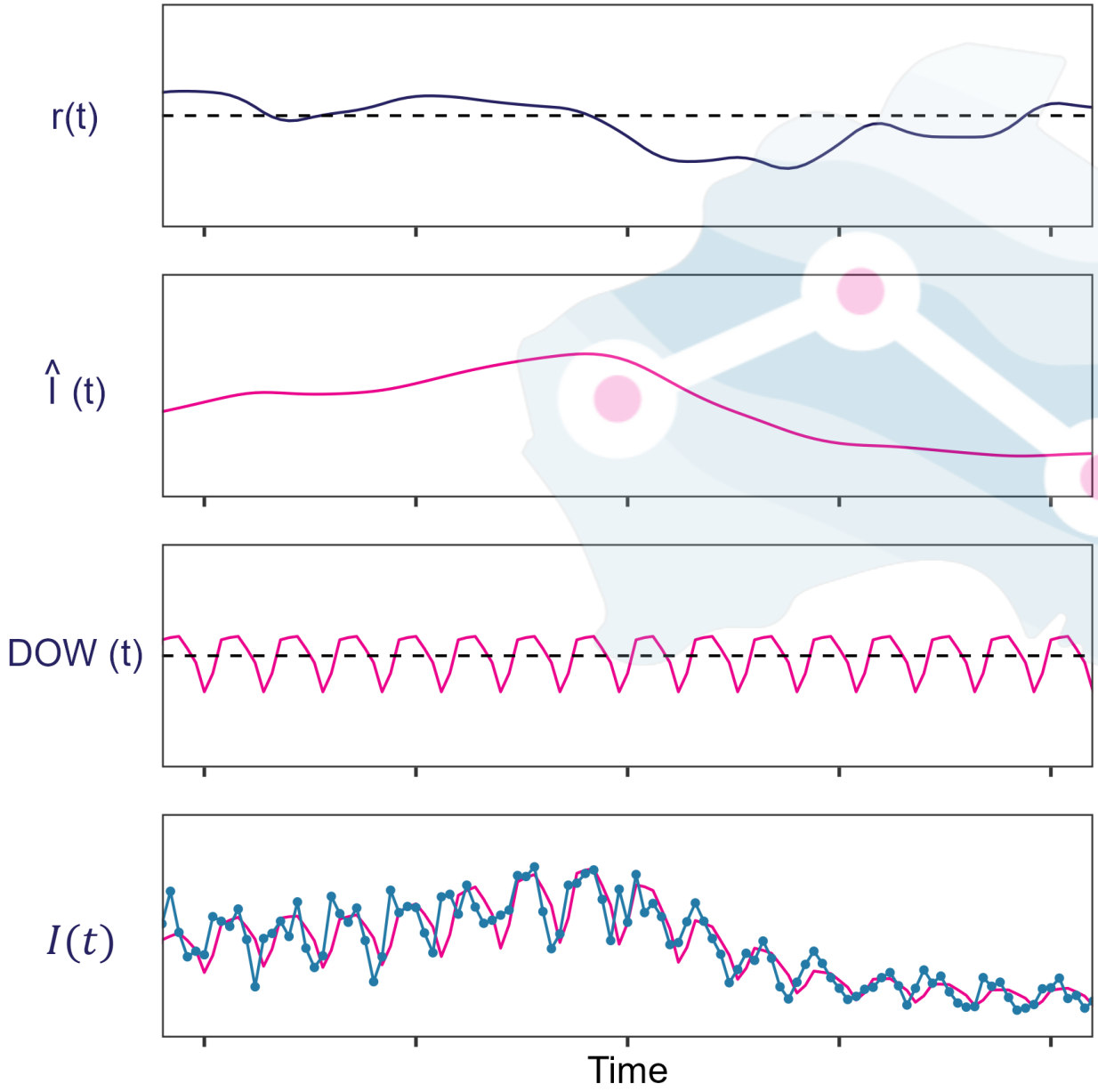
Statistical approaches

- Assume we want to estimate the daily infection incidence, $I(t)$
  - In reality, we observe/have data describing downstream quantities (e.g. cases)
- In the previous lecture, we derived relationships between the infection incidence and the growth rate or reproduction number. For example:



- We can calculate the modelled infection incidence, $\hat{I}(t)$, from the growth rate using the equation above and fit it to data
- We can make assumptions about how $r(t)$ or $R_e(t)$ vary over time.
  - For example, in our next slide we will assume that $r(t)$ varies smoothly over time

Statistical approaches



Smoothed function defined for $r(t)$

Calculate the infection incidence from the growth rate:

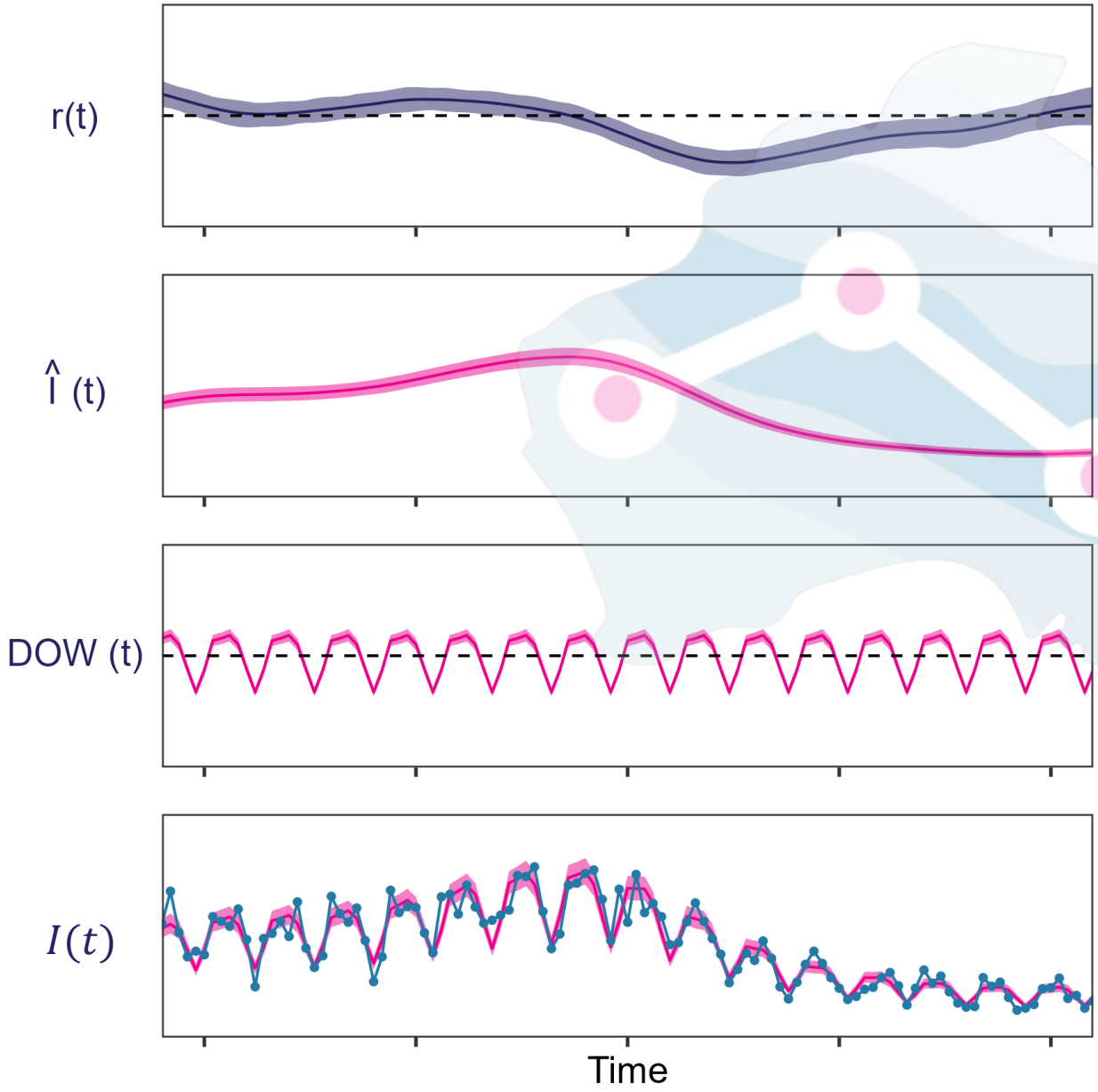
$$\hat{I}(t) = \hat{I}(t-1)e^{r(t)}$$

+

Additional effects: e.g. day-of-the-week effect multiplier $DOW(t)$

Probability of data given modelled values

Statistical approaches



Smoothed function defined for $r(t)$

Calculate the infection incidence from the growth rate:

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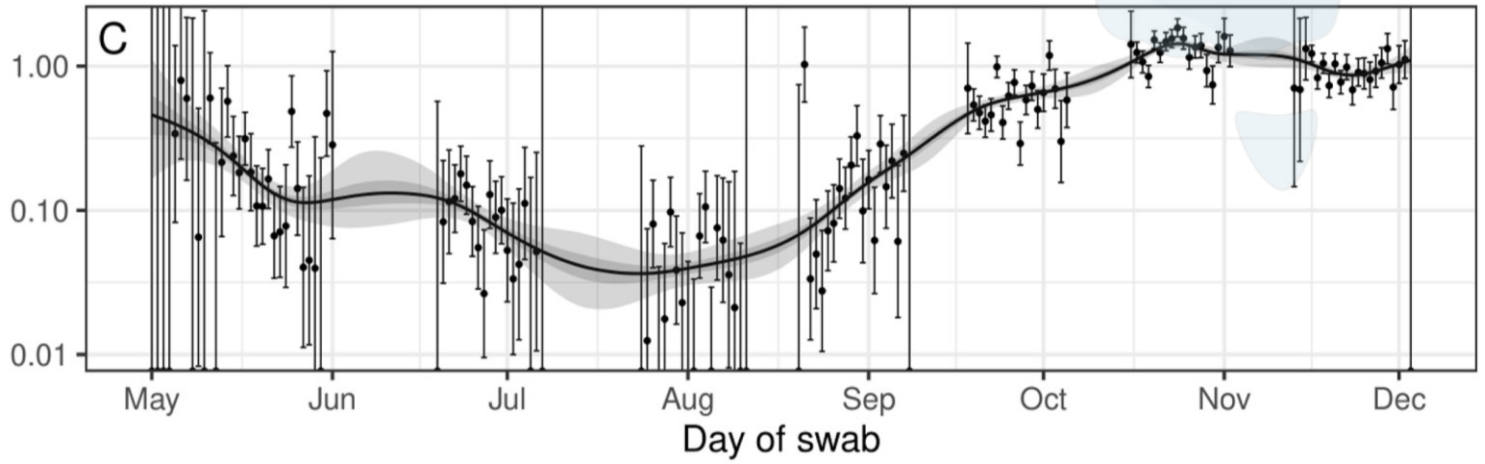
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Additional effects: e.g. day-of-the-week effect multiplier $DOW(t)$

Probability of data given modelled values

Other sources of noise

- There are other sources of noise that can also be accounted for by statistical approaches
  - Data integrity is crucial for allowing appropriate statistical methods to be employed
- For example:
  - Missing periods of data
  - Missing data fields
- It is important when data is missing that it is recorded as such
  - Statistical models can incorporate the additional uncertainty of missing data



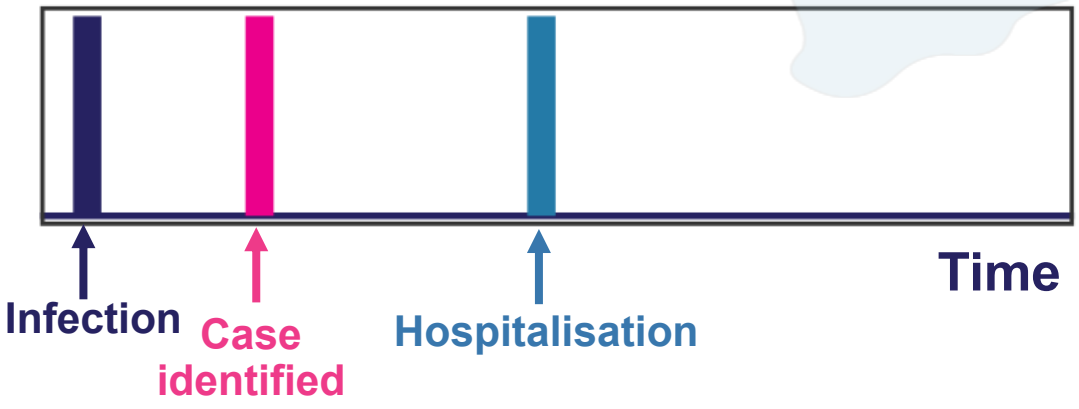
Eales et al. 2022. Epidemics



Accounting for delays in epidemic time series

Downstream outcomes of infection

- Common epidemic time-series (e.g. cases, hospitalisations, deaths) are downstream outcomes of infection
- Not all infections:
  - will be identified as a case
  - result in hospitalisation or death
- The timing of outcomes can vary between individuals



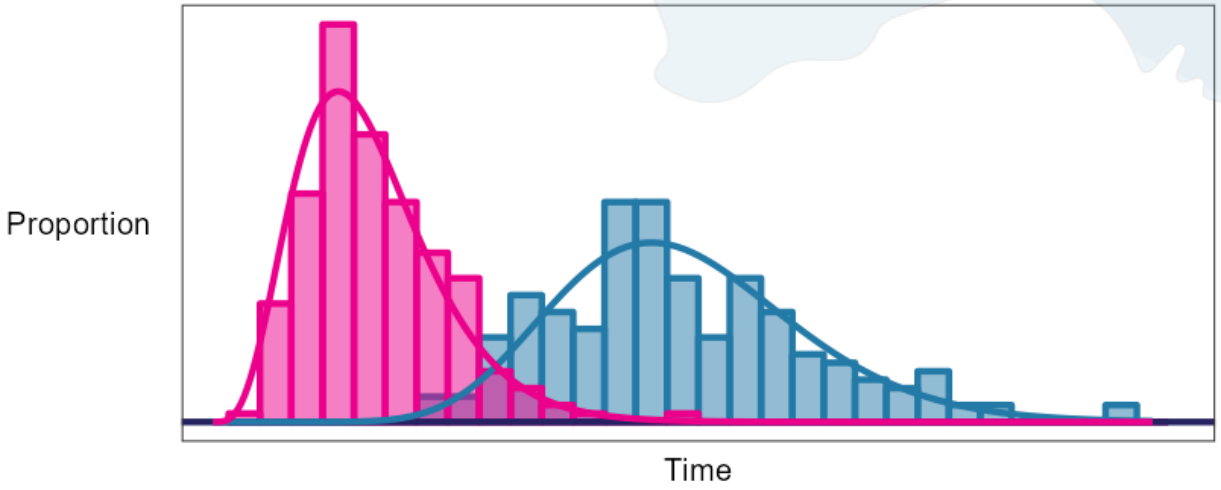


Downstream outcomes of infection

- The time between infection and downstream outcomes can be described by a probability distribution



Event
- Infection
- Case identified
- Hospitalisation



Delay distribution
- Infection to case
- Infection to hospitalisation

Relationship between downstream time series and infection incidence

- We can write the value of a downstream time series in terms of the infection incidence and the probability distribution describing the 'delay distribution', $p(\tau)$

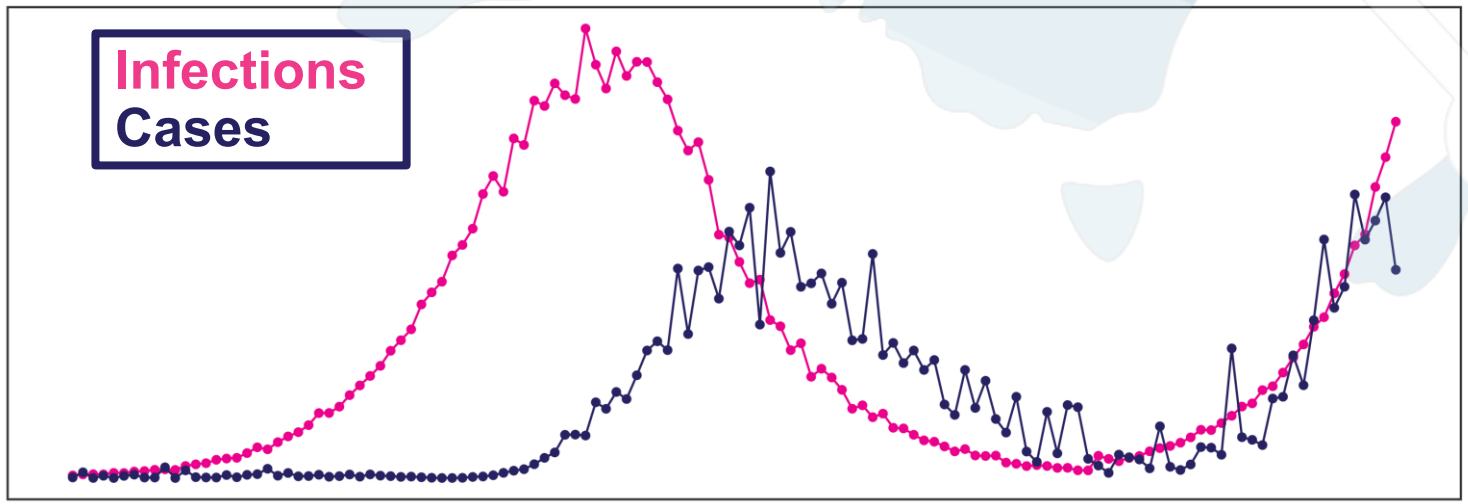
Cases on current day

$$C(t) = \sum_{\tau=0}^{\infty} I(t-\tau)p(\tau)$$

Distribution of delays from infection to confirmed case

Infection incidence on previous days

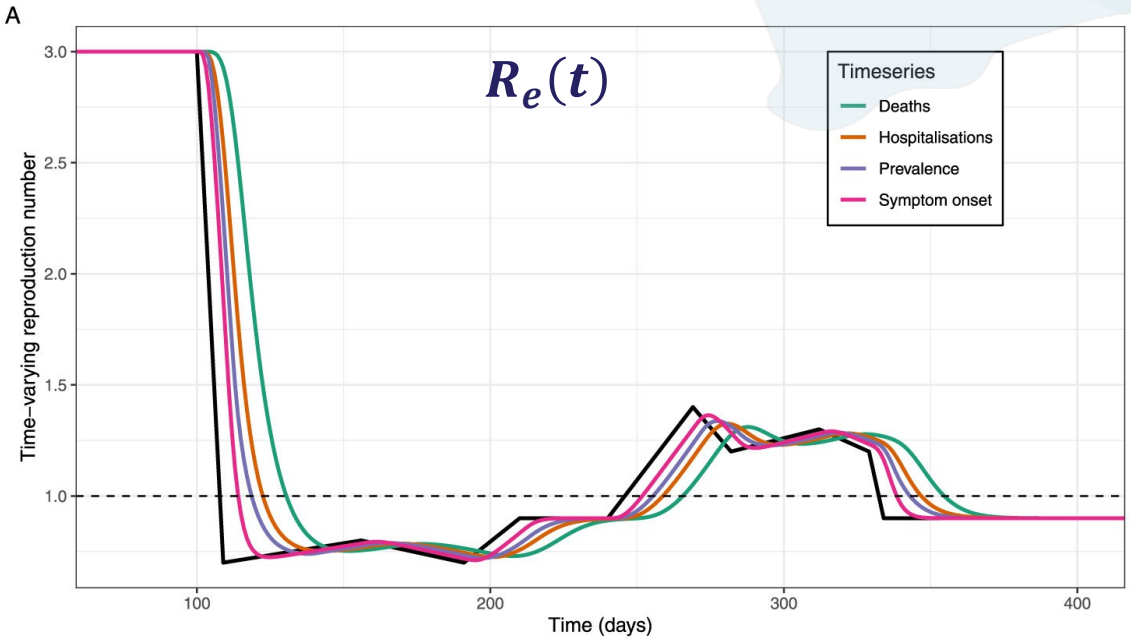
- In general, $C(t)$ does not align with $I(t)$.

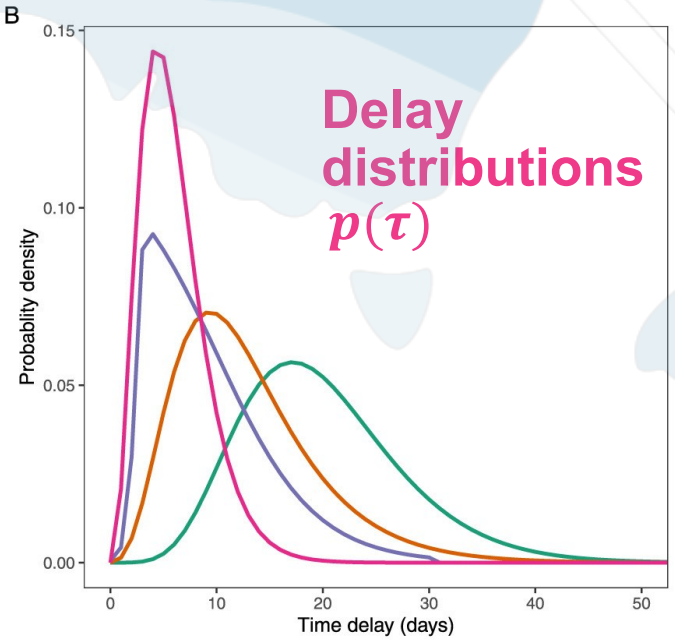


See Eales et al. 2024. Royal Society Open Science for further discussion

Relationship between downstream time series and infection incidence

- Estimates of $R_e(t)$ and $\mathrm{r}(t)$ may be biased if estimated from epidemic time series without accounting for delay distributions:
  - Time from infection to being identified as a case
  - Time from infection to hospitalisation
  - Time from infection to death
  - Time from infection to test positivity (prevalence)



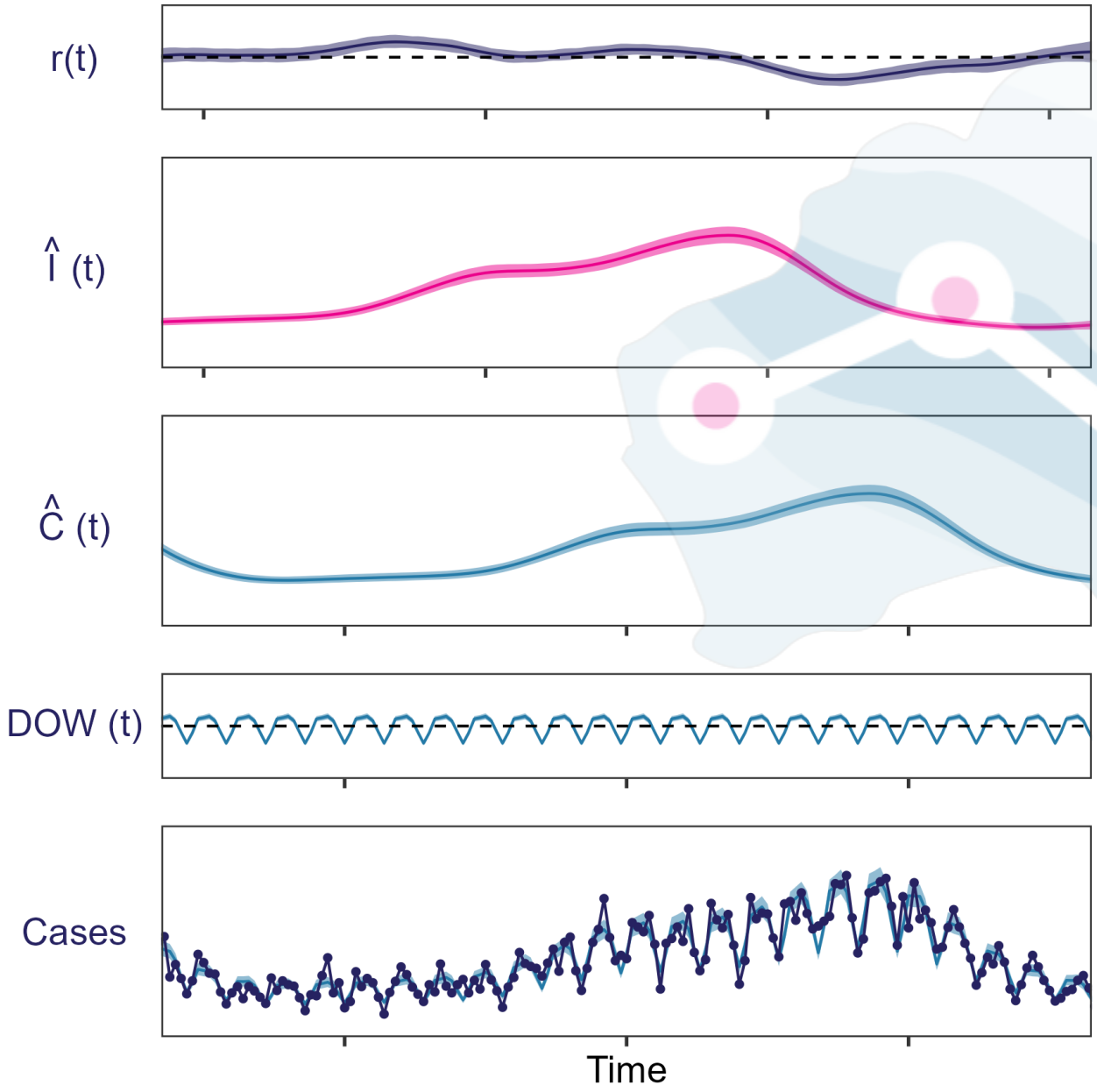


See Eales et al. 2024. Epidemics for further discussion

Methods to account for delay distributions

- Methods exist for estimating $\mathrm{R_e}(t)$ and $r(t)$ while accounting for known delay distributions
- Additional step in statistical models.
- There are multiple R packages that allow for a delay distribution to be provided and accounted for:
  - We will use EpiNow2

Statistical approaches



Smoothed function defined for $r(t)$

↓

Calculate the infection incidence from the growth rate :

$$\hat{I}(t) = \hat{I}(t-1)e^{r(t)}$$

↓

Calculate observed epidemic time series, $\hat{C}(t)$, from $\hat{I}(t)$ using a delay distribution:

$$\hat{C}(t) = \sum_{\tau=0}^{\infty} \hat{I}(t-\tau)p(\tau)$$

+

Additional effects: e.g. day-of-the-week effect multiplier $DOW(t)$

↓

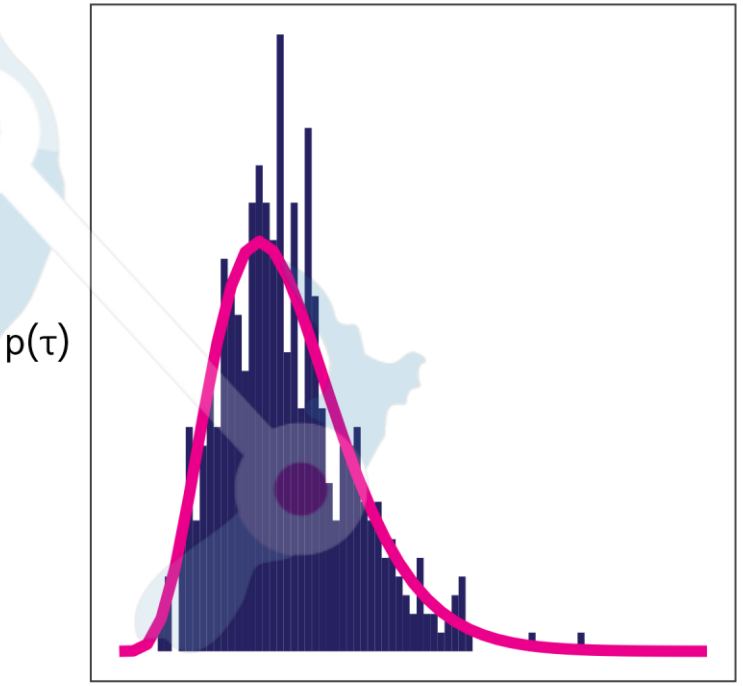
Probability of data given modelled values

Quantifying delay distributions

- To account for delay distributions, we need to know the delay distribution, $p(\tau)$
- Data is often collected that can be used to quantify delay distributions. For example, data describing the time between different outcomes for individuals:
  - Time between symptom onset and case identification
  - Time between identified case and hospitalisation
  - Time between case and death
- Probability distributions can be fit to such data and used when estimating $R_e(t)$ or $r(t)$:
- Published estimates from other countries may also be applicable to the local context.
  - Incubation period (time between infection and symptom onset)

$$C(t) = \sum_{\tau=0}^{\infty} I(t-\tau)p(\tau)$$

Histogram of times from infection to outcome



Biases when quantifying delay distributions

- There are biases to look out for when estimating delay distributions
- The exact time of events might not be known. For example:
  - Time of infection might not be known
  - An individual might not recall exactly when symptoms started
- Missing values or censored values can be taken into account in statistical analyses
- But they must be recorded

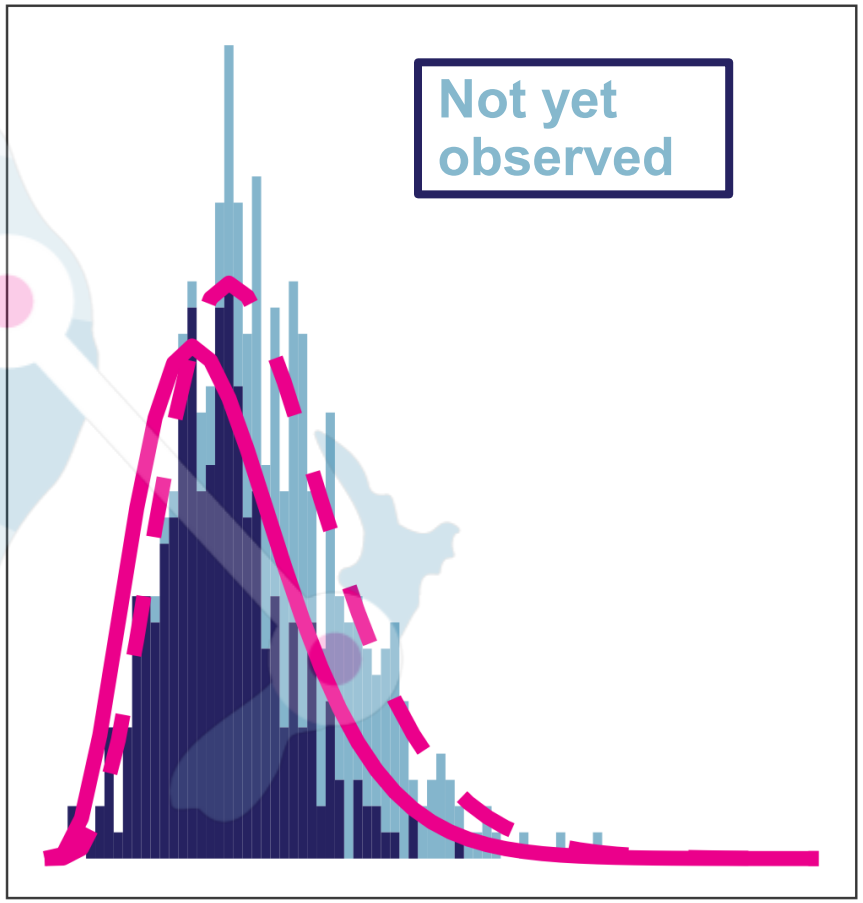
Case number	Symptom onset date (earliest)	Symptom onset date (latest)	Date of case notification
1	2-February-2025	5-February-2025	7-February-2025
2	Not symptomatic	Not symptomatic	9-February-2025
3	MISSING	MISSING	10-February-2025
4	11-February-2025	11-February-2025	12-February-2025

Biases when quantifying delay distributions

- There are biases to look out for when estimating delay distributions
- Longer times from infection to outcomes (hospitalisation, death) are less likely to have been observed i.e. right-censoring.
- This can bias estimates of delay distributions to shorter times.



Histogram of times from infection to outcome



Practical

In today's practical we will:

- Estimate $R_e(t)$ and $r(t)$ using a statistical model (implemented in EpiNow2)
- Compare estimates of $R_e(t)$ and $r(t)$ made:
  - Using different assumed GI distributions
  - Using different datasets
  - Using different datasets correcting for delays

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